

TALOS

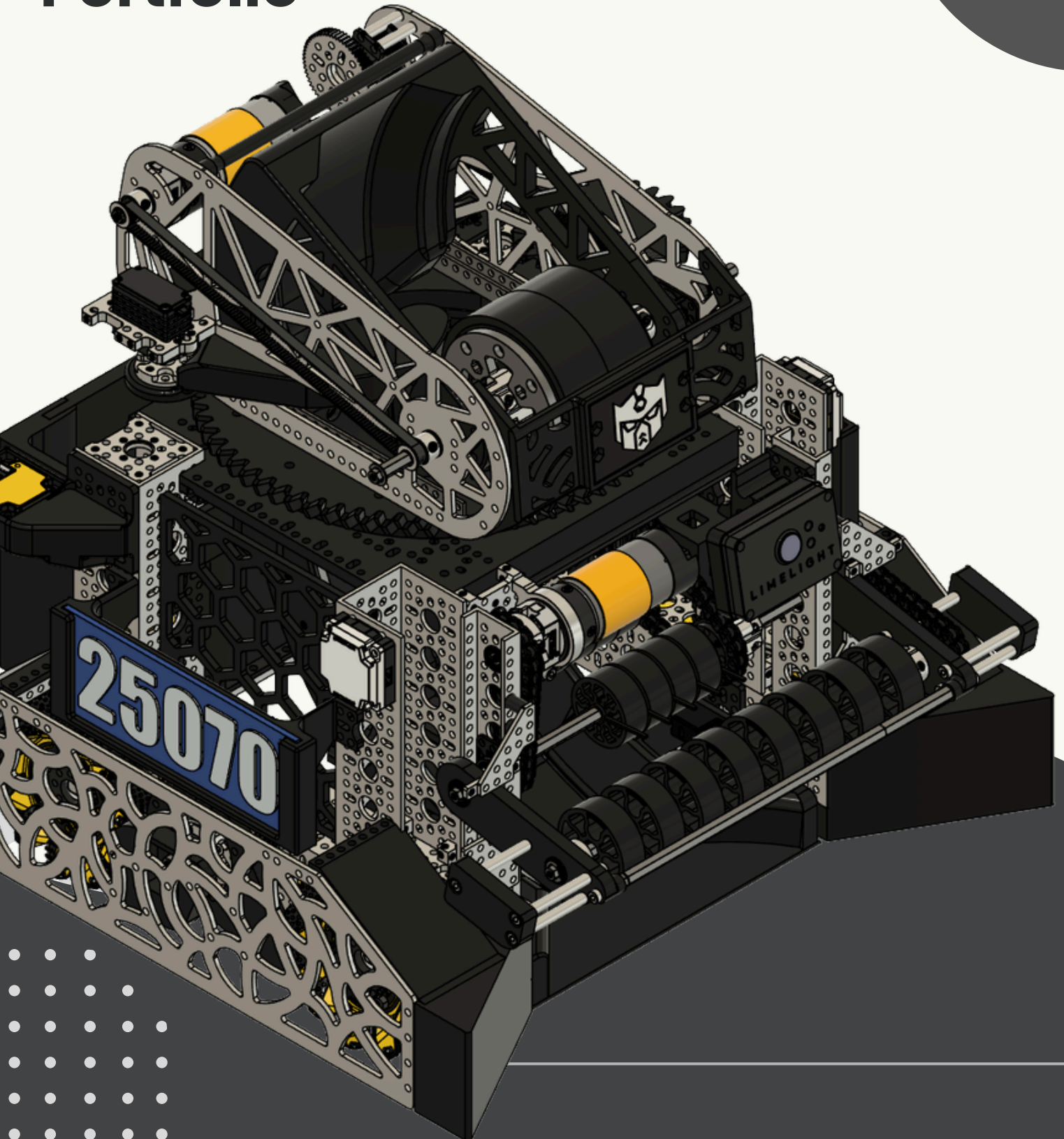
EduRobots

#25070



Engineering Portfolio

2025-2026



INTRODUCTION

1

We are **FTC Team Talos EduRobots 25070**, a student robotics team from **Ierapetra, Crete**, founded in **2023** during the **CENTERSTAGE** season and hosted by **EduRobots**. **Talos**, the **mythological guardian of Crete**, inspired our name. After winning the **Control Award** in Greece and becoming **Finalist Captains**, we qualified for the **Istanbul Premier Event**.

Season Goal: Improve after Nationals and prepare for the 2026 Istanbul Premier Event

MEET OUR MEMBERS

ENGINEERS

Leader



George



Eirini



Manos L.

DESIGNERS

Leader



Stathis



Thanasis

CODERS

Leader



Manos D.



Ioanna

OUTREACH

Leader



Kyveli



Nikos

Our team is organized into **specialized departments**, allowing each member to contribute through their strengths while keeping **communication** and **collaboration** strong across the **entire team**.

AUTONOMOUS STRATEGY ²

Target: 12 artifacts as a team

Four autonomous routines: close and far starts for both alliances.

We decided to create multiple auto programs to accommodate the needs of our **alliance partners**. We're open to **adjusting our routine** to meet our partners' requests if it leads to a higher score.

**Goal for Autonomous →
Build a strong
foundation for the rest
of the match**

CLOSE START FAR START

1

Shoot preload

Shoot preload

2

Collect/shoot 1, 2, 3

Collect/shoot 1, 2

3

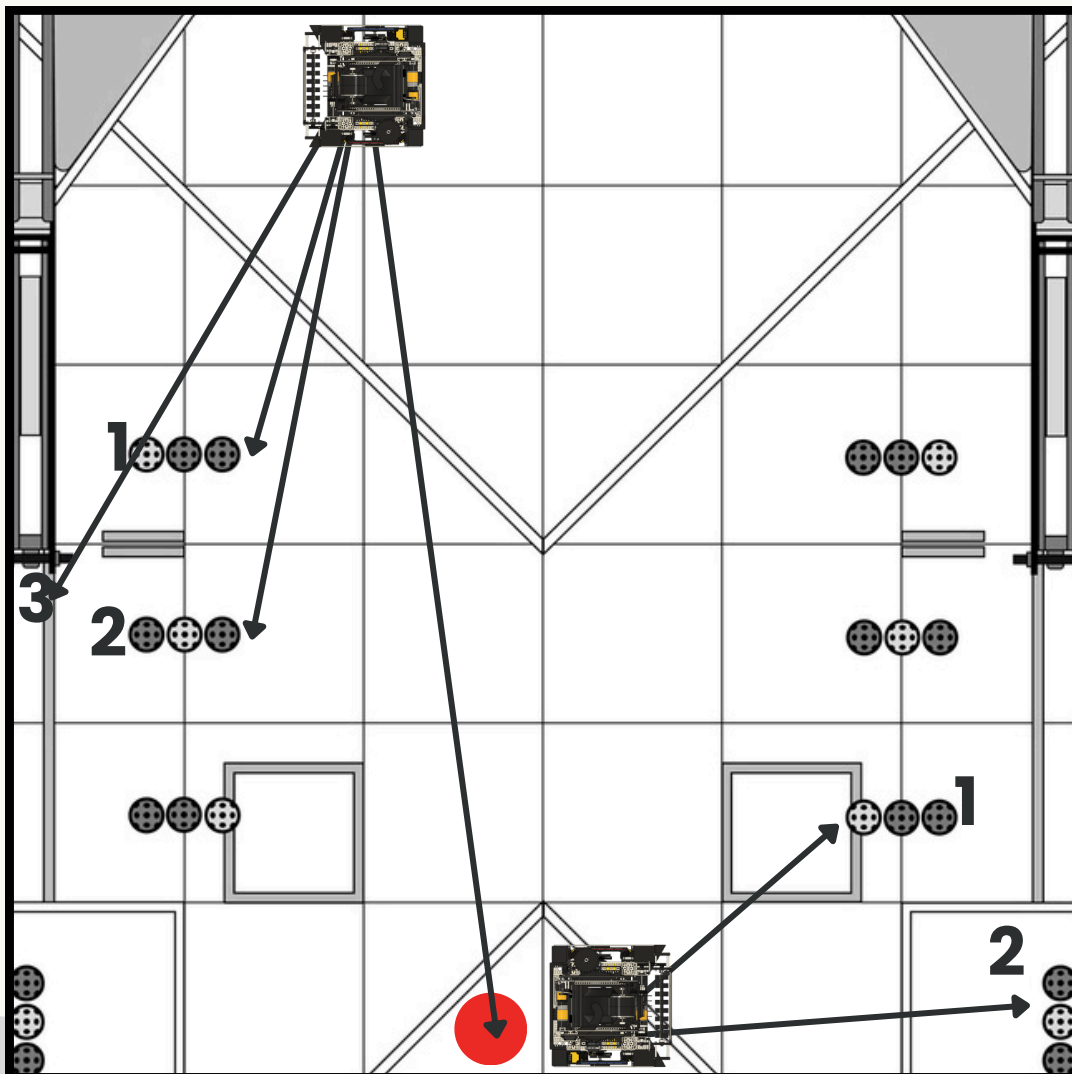
Park at



Park at



CLOSE START



FAR START

TELEOP STRATEGY

3

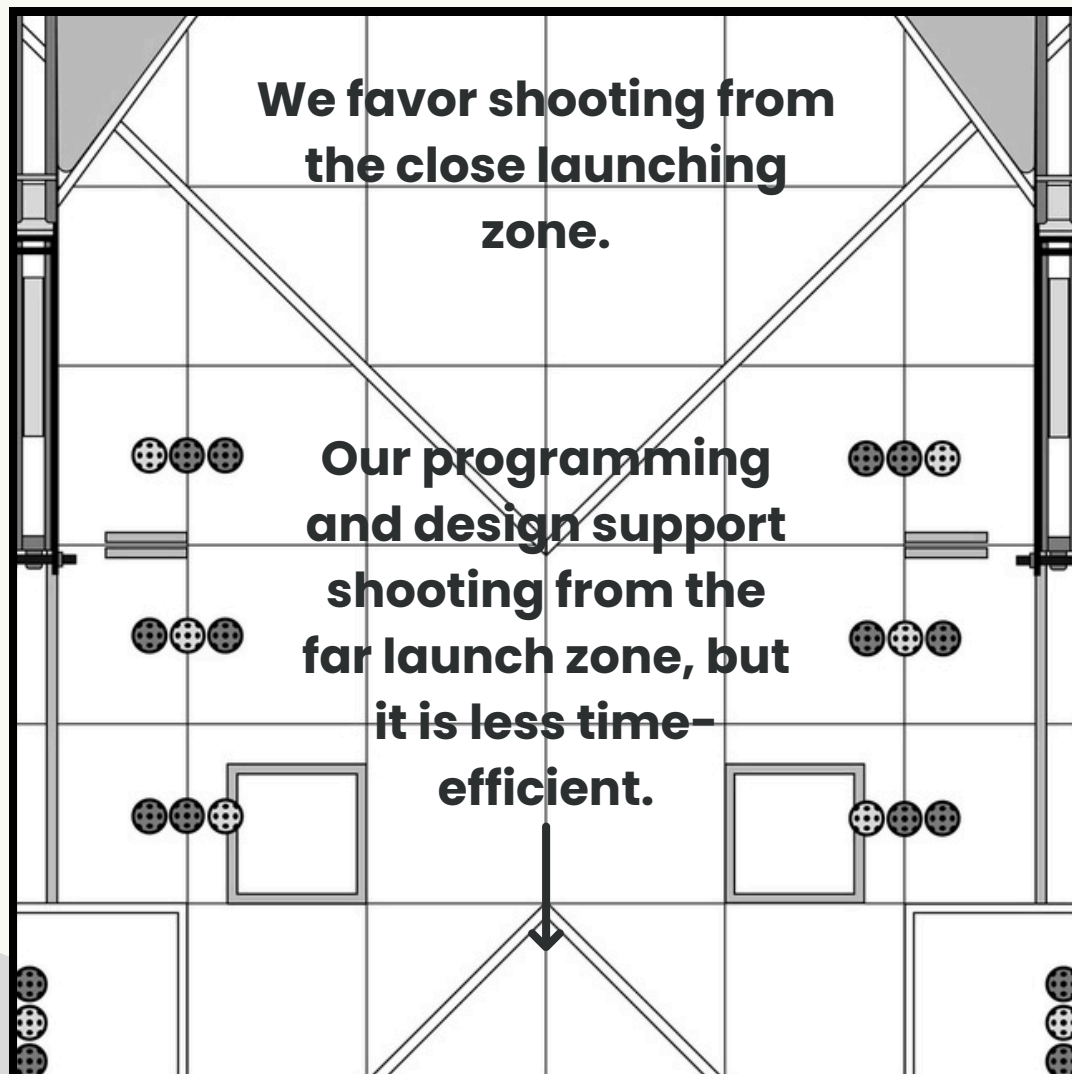
While competing at **FTC Greece**, our strategy was aiming for the **Pattern Ranking Point**, but at the competition, this proved our strategy aimed for the Pattern Ranking Point was impractical for us, because the **complexity** required to earn that RP took **too much time**

Our main strategic priority was securing the **Goal Ranking Point** through fast, reliable artifact scoring. Our game analysis showed that **rapid shooting** offered the best balance of **points, consistency, and alliance value**, so we designed the robot for quick intake, stable transfer, precise shooting, and adaptable autonomous routines.

Target: 36+ artifacts per match with our alliance

We are careful to:

- Have **fast cycle times**
- Focus on the **number of artifacts**, not the pattern
- Have good communication with **our alliance partners** to avoid fouls and miscommunication



We are also careful with our **timing** so that we can **avoid conflict** with **opponents** caused by our own timing or positioning mistakes.

If we cannot score efficiently, we **focus on defence**.

At the end of Teleop, we also aim to score the **Movement RP** by using our **lifting mechanism** and parking with our partner.

DESIGN PRIORITIES

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How Strategy Became Robot Requirements:

Strategic Need	Requirement	Robot Feature
Score many artifacts quickly	Fast cycle time	Active intake & Rapid transfer
Reduce missed shots	Accurate aiming	Turret, hood, & velocity control
Score in auto	Repeatable autonomous	Mirrored close & far routines
Adapt to partners	Flexible paths	Multiple autonomous programs
Reduce failures	Strong mechanisms	Redesigned transfer system, & gear control
Reduce cycle time	Easy driving	Compliant intake & automated targeting

Strategic Tradeoffs:

We chose **speed and repeatability** over unnecessary complexity. Although the **Pattern Ranking Points** are valuable, our most reliable contribution was **consistently scoring many artifacts**. This decision shaped every major subsystem: the intake had to be forgiving, the shooter had to return to target velocity quickly, the turret had to aim consistently, and the autonomous had to work from multiple starting positions.

**Pattern
Perfection** 

**Speed &
Quantity** 

**Complex
Mechanisms** 

**Reliable &
Consistent** 

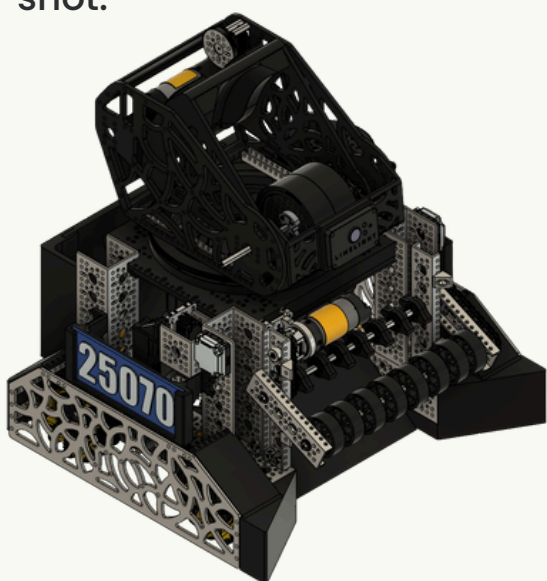
**Fixed Auto
Routine** 

**Alliance
Flexibility** 

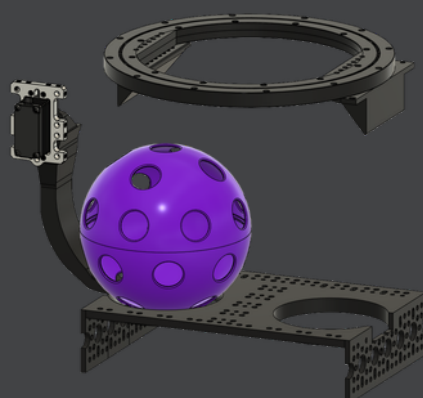
DESIGN EVOLUTION

V1 → Flicker-assisted robot

The artifact enters the robot via the **Active Suspension Intake**, driven by a motor whose motion is transmitted via a chain. It moves up the **ramp**, aided by the intake, and enters one of two **round openings that connect to the flickers** or **remains staged in the intake**. The flickers (powered by servo motors) move the artifact to the shooter in a “flicking” motion, then the turret and hood aim the shot.

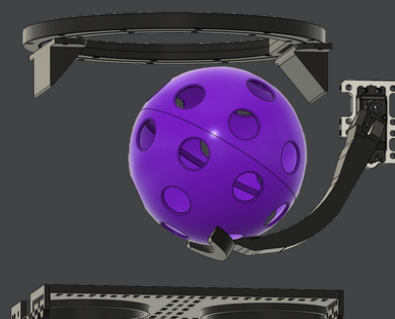


Position 1



The base supports the artifact

Position 2



The flicker leads the artifact to shooter.

We identified that our main problem after Nationals was **inconsistent transfer and shooting**. We tested multiple solutions, **compared tradeoffs**, redesigned the **transfer/turret/shooter**, and verified improvement through **shooter velocity, turret error, and shot consistency data**.

Biggest issues: Speed+Consistency

SYSTEM	BENEFITS	OBSTACLES	CONSEQUENCE
Flicker Mechanism	Moved artifacts the toward shooter	Misfeeds or inconsistent artifact position	Missed shots or delays
Shooter	Could score when fed correctly	Compression and feeding inconsistency affected aim	Lower repeatability
Turret	Allowed aiming	Gear skipping and not aligning	Slower aiming, mechanical stress

DESIGN EVOLUTION

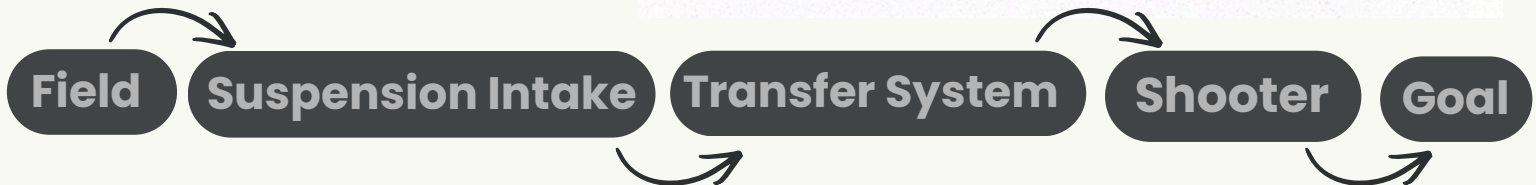
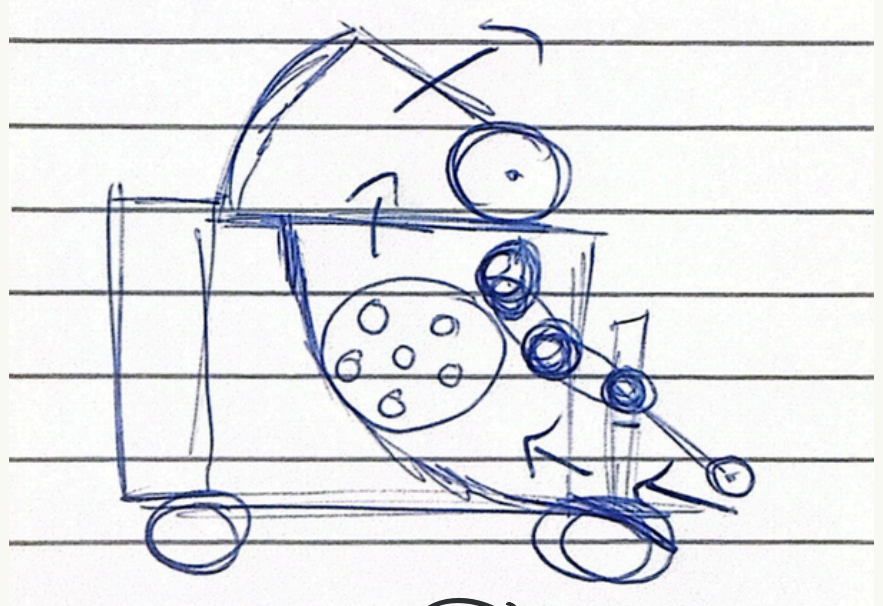
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V2 → Active Transfer System

First sketch of the new robot

We had to change major mechanisms: the transfer system was the biggest alteration, and then the shooter and the turret. With these changes, we aimed to achieve:

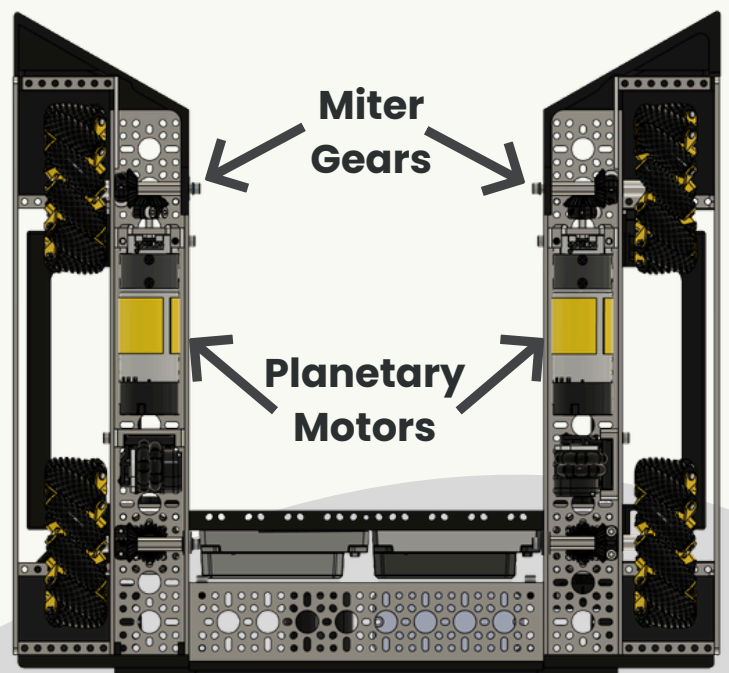
- **Faster cycles**
- **More consistent feeding**
- **Less driver error**
- **More forgiving artifact movement**
- **More elegant and accesible**



SUBSYSTEM ANALYSIS

Drivetrain

- **Compact mecanum drivetrain**
- **Four 312 RPM planetary motors** mounted inside or above the goBILDA channel frame
- **GripForce mecanum wheels** provide strong traction on **FTC foam tiles**
- **Steel miter gears** and 8mm shafts **transfer power** reliably in a space-efficient way.



This setup lets us **drive, strafe, and rotate smoothly** while keeping the chassis low-profile and **easy to modify**.

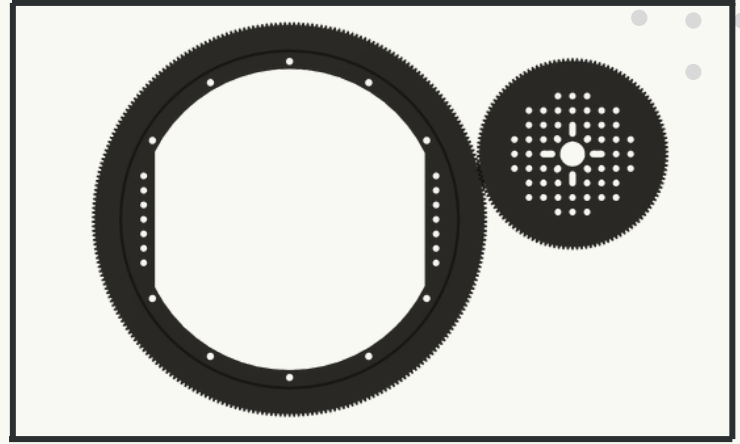
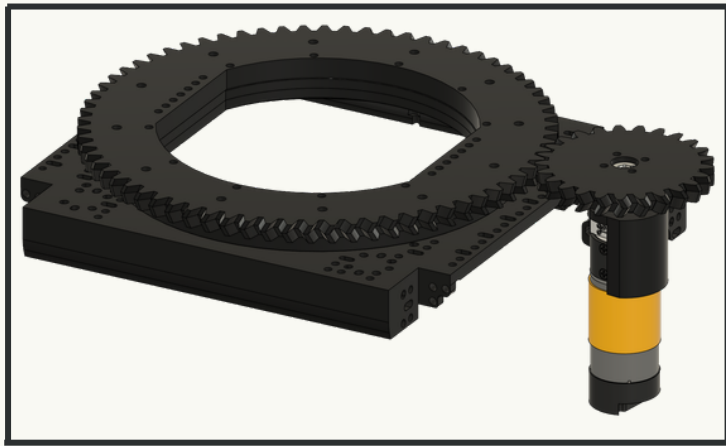
SUBSYSTEM ANALYSIS 7

Turret

We used a **turret** to reduce **driver alignment error** and improve **shooting accuracy**, allowing the driver to focus on positioning and strategy.

Pre-Nationals

Before competing, our turret was a simple, **small-tooth, 3D-printed, 260-tooth gear** driven by a smaller 130-tooth gear powered by a 312-RPM motor. Gear reduction was **2:1**, and the **final RPM was 156**. The gear was connected to a **bearing** that aids in smooth movement.

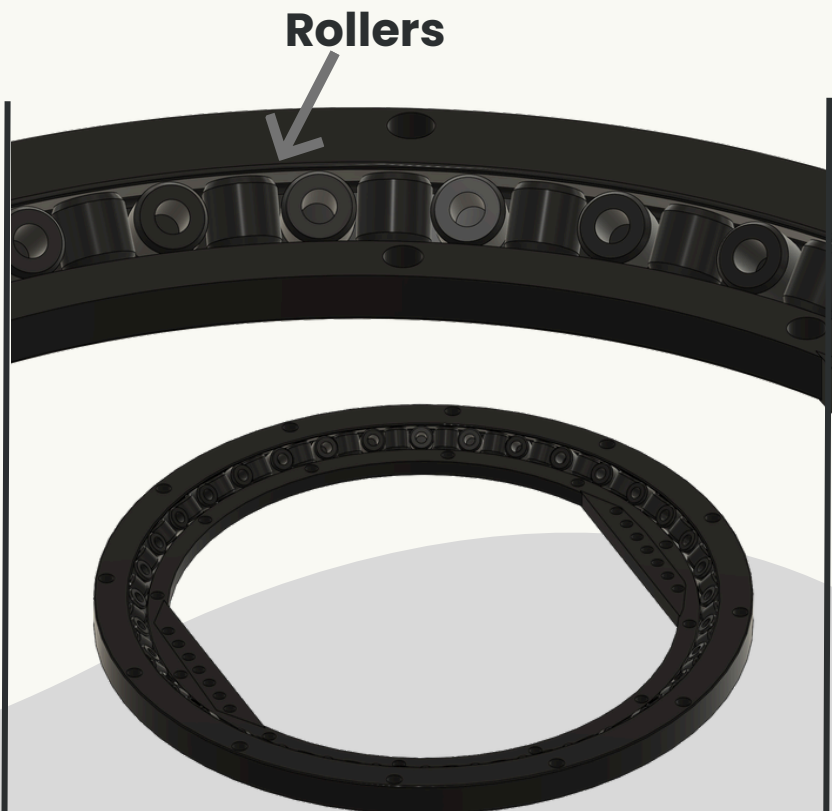


Post-Nationals

After the competition, we replaced the basic gears with **innovative 3D-printed herringbone gears** to **reduce skipping and improve consistency**. The turret now uses a **70-tooth gear driven by a 25-tooth gear** on a 312 RPM motor, changing to a **2.8:1 gear reduction and 111.4 final RPM**.

Bearing

The turret is supported by a **bearing** for smooth, rapid rotation. This is a 3D-printed mechanism that has an **83.5mm inner diameter and a 216mm outer diameter**. Because it is fully 3D-printed, we use **rollers turned 90 degrees from each other**, making it easier to print, smoother, more efficient, and lower friction. The roller has an **8mm outer diameter and 7.5 mm length**.



SUBSYSTEM ANALYSIS 8

Lift

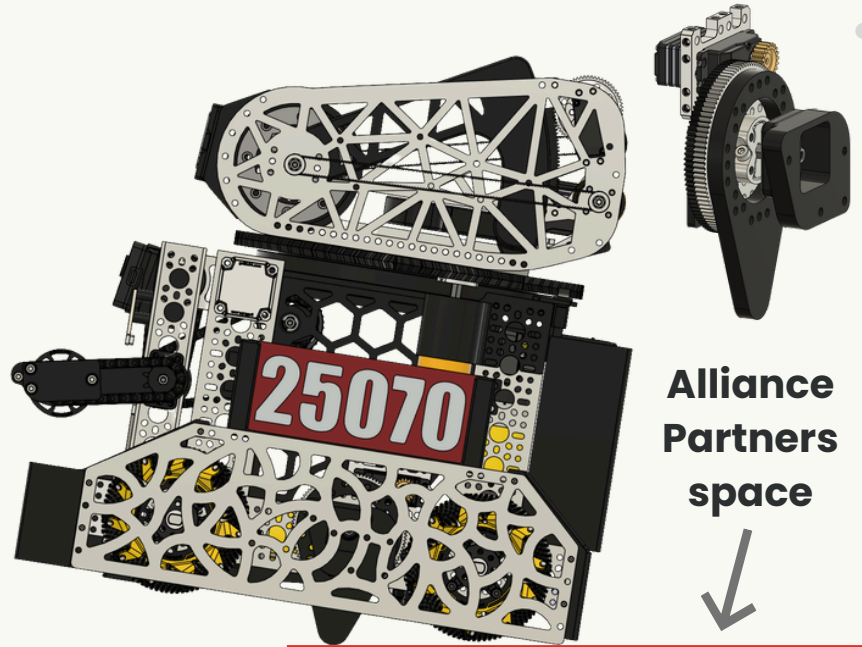
For the lift, we chose a simple one, so we could focus time on **higher-priority mechanisms**. Using **3D-printed supports**, the robot **lifts at an angle** while keeping some wheels on the ground, leaving enough base space for our **alliance partner** to park.

Pros:

- Allows alliance partner parking
- Stable & Reliable.
- Simple, so no time lost from more important mechanisms

Cons:

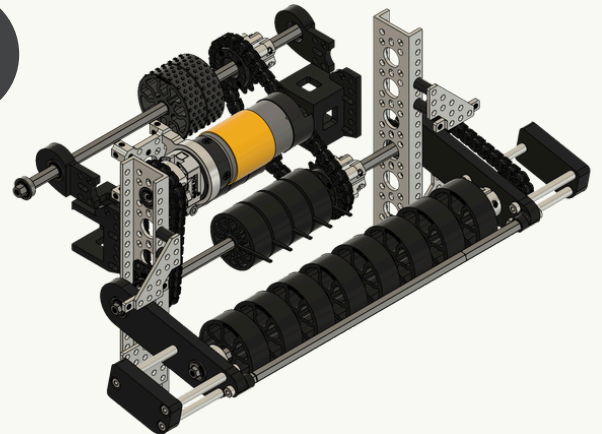
- Risk of alliance partners or opponents tipping the robot
- Does not always allow two robots to fully park



Pros > Cons

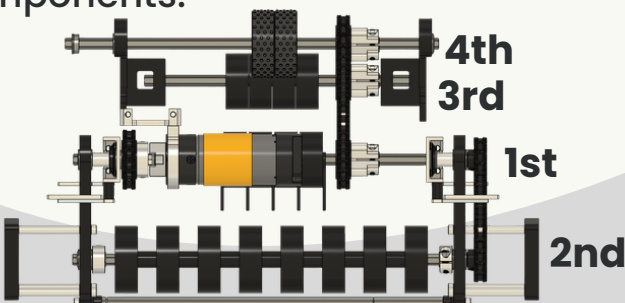
Intake & Transfer

Although we refer to these two systems as separate parts of the robot, they are **interconnected**. One cannot work without the other. The system has **four rows: two for the intake and two to lead the artifact to the shooter**. They are made up of an **8mm shaft** with different components.



Intake:

- Driven by a **1150 RPM motor** geared **1:1.4** overdrive → **Final RPM 1610**
- One **stationary shaft** with **40mm diameter wheels** and **zip-ties**
- One **suspended shaft** with **48 mm diameter wheels** and movement transferred by chain. **Ratio 10:10 (1:1)**

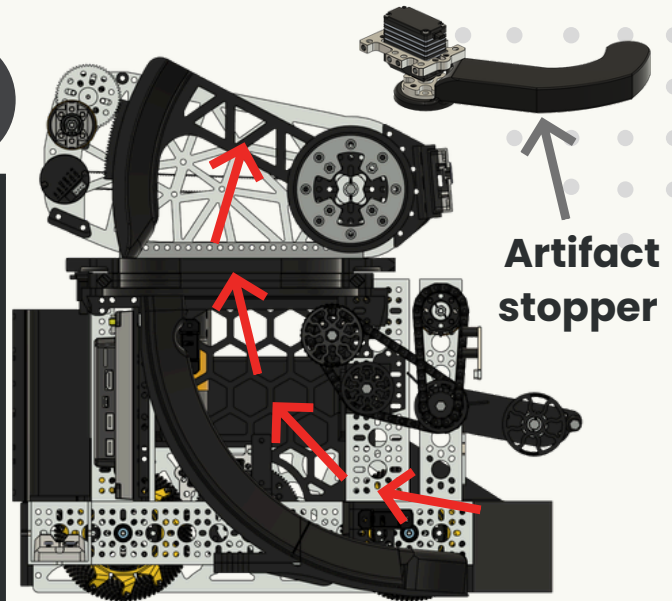


SUBSYSTEM ANALYSIS 9

Intake & Transfer

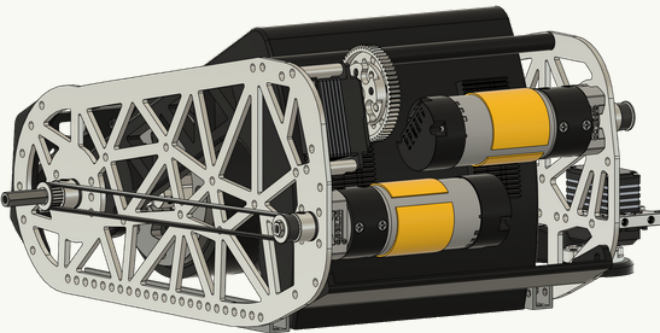
Transfer

- Driven by the same **1150 RPM motor** geared **1:1.4 overdrive** → **Final RPM 1610**
- Two **stationary shafts**, both geared **14:14 (1:1)** and motion transferred via **chain**
- **First shaft** → **48mm** diameter wheels
- **Second shaft** → **40 mm** diameter wheels
- Artifact stopper powered by a servo motor



Shooter

For the shooter, we chose a **flywheel system** for its **simplicity and speed**. To avoid **manual repositioning**, we added an **automated hood** to adjust the shooting angle.

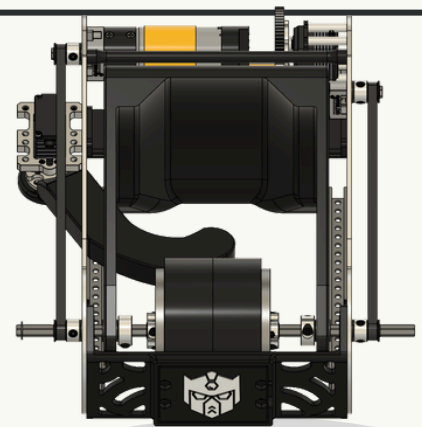


Launcher

- **Dual-motor-powered flywheel shaft 6000 RPM**
- **2mm gt2 timing belt, ratio 1:1**
- Two **steel goBILDA flywheels** for more **moment of inertia (82mm diameter, 152g, 1651 g·cm²)**
- Two goBILDA 92mm diameter **Rhino wheels**

Hood

- Powered by a **servo** to change the **angle of the shoot**
- Compression → **2mm 127mm artifact**
- **4mm of 125mm artifact**, lightly deformed
- goBILDA compatible gear for **adjustable hood angle**



Our robot is efficient for one clear reason: it is **simple**. Because of its simplicity, it is **elegant and robust**, leading to **easy repair and fast changes**.

Our robot matches our **strategy** and our **drivers' preferences**, leaving us with a combination that not only meets designers' priorities, but **leaves every department satisfied**.

Code Architecture & Driver Automation

We focused on **reducing the number of decisions the driver must make** during a cycle. The driver chooses the strategy, but the robot handles **repeated timing-sensitive actions**: aiming the turret, changing **shooter velocity** and **hood angle**, stopping intake when full, and using **LEDs to show different robot states**.

Control Flow

Pinpoint Odometry + Fixed Limelight April Tag Observation

→ **Validated pose relocalization** → **Corrected Position**

Corrected robot pose → **Turret Position LUT** → **Virtual aim point**

→ **Profiled turret controller** → **Distance LUT** → **Shooter Velocity**

→ **PIDFF flywheel controller** → **Distance + Measured velocity LUT**

→ **Hood Position**

3 distance sensors + Intake current → **Transfer state machine**

→ **Shooter/Turret/Transfer State** → **LEDs + Panels telemetry**

LED Status	Robot State
flashing green	intake state
orange	robot empty
purple	shooter velocity not yet at target velocity
static green	ready to shoot

Intent	Robot Response
Drive	Field-centric drive with acceleration limiting; clicking the stick enables 25% slow mode
Collect	Transfer enters COLLECT; the intake stops automatically when full or when overcurrent is detected
Shoot	Holding the bumper opens the gate and runs the transfer
Localize	A fixed Limelight above the intake uses AprilTags to correct accumulated odometry drift
Aim	Turret continuously calculates its angle from corrected robot pose and the interpolated virtual aim point
Select fallback	The driver can use a face-forward override or fixed close/far values if the LUT system fails or is disabled
Read status	LEDs show shooter/target and transfer states; telemetry exposes detailed debug values

Earlier robot	Current robot
Driver timed multiple flickers and watched artifact position	One transfer state machine controls collection, stopping, and feeding
Two fixed velocity/hood combinations	Velocity and hood change continuously from pose and measured shooter velocity
Turret camera could lose the target while rotating	Fixed Limelight relocalizes the pinpoint odometry computer while pose/LUT aiming controls the turret
Turret aimed at one fixed field target	Position LUT changes the virtual aim point across the field

Code Architecture & Driver Automation

TURRET

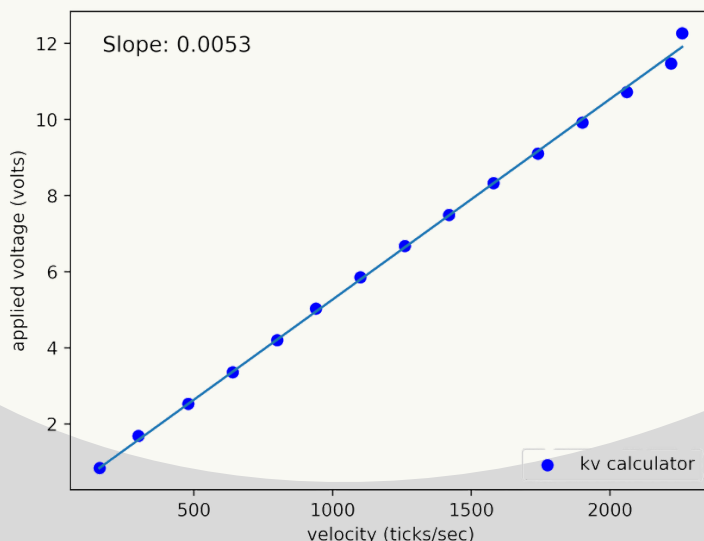
Version	Main idea	Why we changed it
V1	Limelight mounted on the turret directly measured the target error	Lost target with turret rotation, stressed mechanics, and still had poor aim from some field positions
V2	Pinpoint pose plus trapezoidal motion profiling aimed to one fixed field target	Removed dependence on camera FOV and mechanical stress, but odometry drift and one fixed target still limited shooting accuracy
V3	Limelight relocalization plus position aim LUT	Corrected odometry drift and changed the virtual aim point across the field

Shooter Controller

$$\text{power} = \text{PID}(v_{\text{target}} - v_{\text{measured}}) + \frac{k_s \text{sign}(v_{\text{target}}) + k_v v_{\text{target}}}{V_{\text{battery}}}$$

Encoder feedback corrects velocity error, while voltage-scaled feedforward predicts the power needed to hold speed before error grows.

We tuned the **Kv constant** by finding the **slope of a logged voltage-velocity linear regression**. Then we verified the result with **closed-loop error and recovery graphs**.

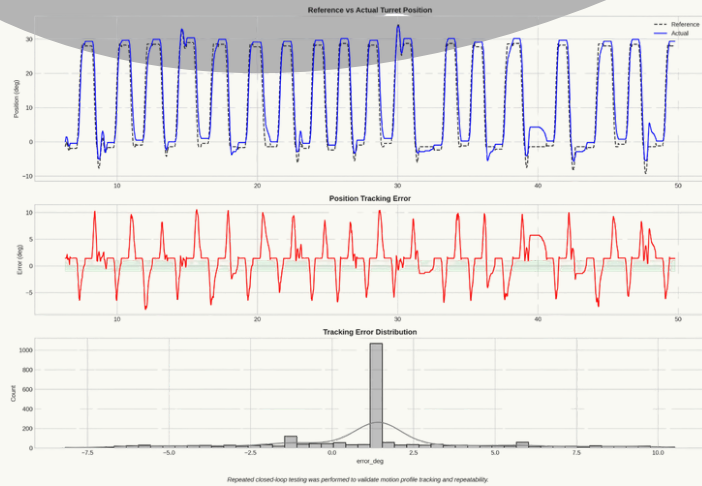


System	Input / Feedback	Algorithm	Output
Turret	Corrected field pose + turret encoder	2D inverse-distance LUT, geometry, trapezoidal profile, PIDFF	Turret angle
Shooter	Corrected goal distance + flywheel encoder + battery voltage	Linear distance LUT + velocity PIDFF	Flywheel velocity
Hood	Corrected goal distance + measured flywheel velocity	2D inverse-distance lookup table (LUT)	Hood servo position
Transfer	Three distance sensors + intake current	STOP, COLLECT, SHOOT state machine	Intake and gate commands

DATA & TESTING

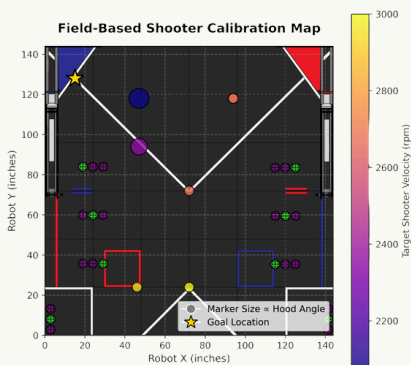
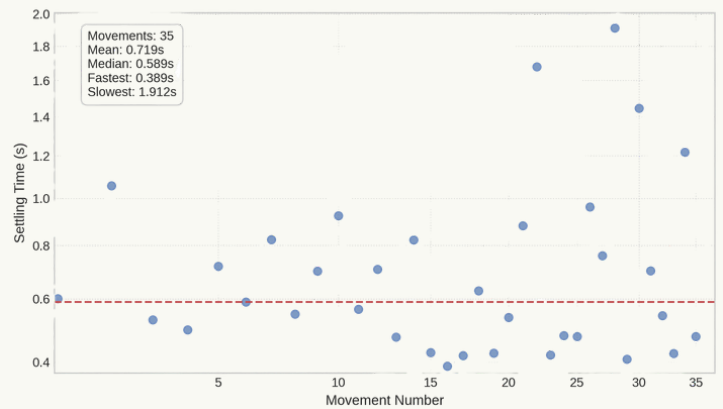
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Turret Motion Profile Validation and Performance Characterization

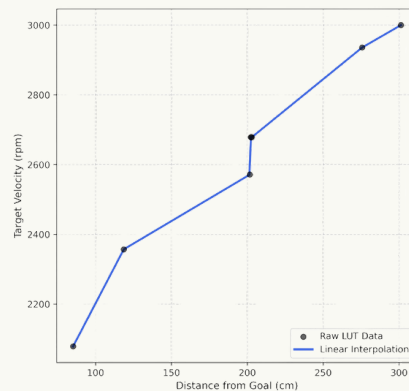


This graph shows the **turret's settling time** for **each repeated movement during automated testing**. Each point represents the **time required for the turret to reach and stabilize at its target position within the allowed tolerance**. By analyzing many consecutive movements, we verified that the **motion profile and controller produce fast and repeatable performance**, ensuring the turret is **ready to shoot reliably during a match**.

Turret Settling Time During Repeated Testing



Shooter Velocity LUT with Linear Interpolation



Shot data was collected from **multiple locations on the field** to determine the **optimal shooter velocity and hood position** for each distance. The field map shows where **calibration data** was gathered, while the **LUT graph** shows how these measurements are converted into a continuous model using **linear interpolation**. This allows the robot to **automatically calculate shooter settings** for any position on the field, improving shot consistency and reducing manual tuning.

This graph compares the **shooter's response when using a PID + Feedforward controller versus a fixed-command**. By examining spin-up time, overshoot, and settling time within the target velocity tolerance, we verified that the **closed-loop controller reaches the target speed faster and with greater accuracy**, resulting in **more consistent shooter performance**.

comparison: PID + FF Controller vs Open Loop Power Level: Time vs Shooter Velocity for PID + FF and Open Loop Power



REACH & CONNECT 13

This season, we achieved:

- 19+ events
- 10.000+ people reached
- 20+ hours mentoring
- 2 international events

Our philosophy is built on:

Active participation

Inspiring people

Accessability

Inclusivity

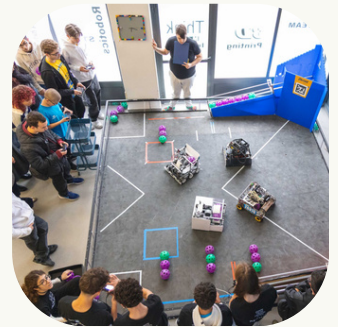
Meeting with the Women Do Business organization

Met with Women Do Business to discuss **leadership, equality, and women's empowerment in STEM and robotics.**



Cretan FTC scrimmage event 2026

Hosted the first **Cretan Scrimmage**, giving FTC teams a chance to **practice, improve, and collaborate** before the National competition.



FTCxLive seminar 2026

Launched the **FTCxLive seminar**, to promote **collaboration, learning, and connection within the international FTC community.**

Participation was open to all FTC teams.



2nd Experimental High School visit

Presented our **FTC journey**, showcased our robot, and introduced students to **robotics, STEM, and the values of FIRST** through an interactive presentation.



REACH & CONNECT 14



AI Seminars with Paulius Uza

Self-taught entrepreneur, founder of InRuntime, co-founder, designing innovative blockchain-powered tools, former CTO of BDSwiss

STEM presentation at high school computer science class

8/05/26

STEM and archeology presentation at a local kindergarten.

6/02/26

Workshops with FLL Team **CDG #C3135** (mechanics, CAD and programming)

12/10/25

Assistance to FTC Team **Robonagers #28870** (equipment and field material)

**12/1/26
-13/3/26**

Assistance to FTC Team **Hex Wizards #33512** (field material)

27/2/26

Assistance to FTC Team **Pandurata 26523** (mentorship-focused sessions)

**5/02/26
-10/3/26**

Mentoring FTC Team **Leocopsals #26487**

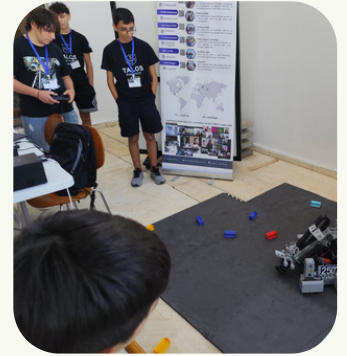
**20/10/25
-10/3/26**

Mentoring FLL Team **CDG #C3135**

**1/10/25
-13/3/26**

FORTH Researchers' Night 2025

4k+ attendees of all ages



Total Lunar Eclipse Telescope Night 2025

2k+ attendees of all ages



FLL Heraklion volunteering 2026

1k+ attendees of all ages



AI-driven apps to improve portfolios and presentations



Apps that let teams practice and perfect their portfolios and judging presentations.

SUSTAINABILITY

15

FUNDING

25.282€

Eugenides Foundation	3.000 €
Holy Metropolis of Ierapytna and Siteia	3.000 €
Hellenic Parliament	2.000€
Ierapetra Municipality	1.750 €
Plastika Kritis	1.500€
Angelicooussis Foundation	1.500€
Eurobank	1.000€
Gene Haas Foundation	500€
Other Companies	11.032 €

MEMBER DEVELOPMENT

Internal team seminars

Experienced members help new members with:

- Programming
- CAD
- Engineering
- Outreach



EXPENSES

24.950€

FTC Registration EuPE25, GrNC, EuPE26	3.875€
Equipment	6.000€
Airfare Tickets	2.710€
Hotel	7.365€
Meals and Travel Transfers	5.000€

Pre-FTC Team

A year-round program for students interested in joining our team. The goal is to teach them all aspects of FTC.

Session with members of CDG



Sponsorship agreements

Our one-year sponsorship agreements are organized into Partner, Gold, Silver, Bronze, and Supporter tiers. Each tier offers benefits matched to the sponsor's goals and capacity.

PARTNERS:



ΙΔΡΥΜΑ ΕΥΓΕΝΙΔΟΥ



HELLENIC PARLIAMENT



ΜΟΡΦΩΤΙΚΗ
ΣΤΕΓΗ
ΙΕΡΑΠΕΤΡΑΣ

ΔΗΜΟΣ Ierapetra.gov.gr
ΙΕΡΑΠΕΤΡΑΣ

XSTEAM
EDUCATION

ΙΔΡΥΜΑ
ΑΝΤΩΝΙΟΣ & ΙΩΑΝΝΗΣ
ΑΓΓΕΛΙΚΟΥΣΗΣ



ΝΕΟΣ ΠΟΛΙΤΙΣΤΙΚΟΣ
ΣΥΛΛΟΓΟΣ ΙΕΡΑΠΕΤΡΑΣ

GOLD:



ΠΛΑΣΤΙΚΑ
ΚΡΗΤΗΣ Α.Ε.



iqcrops.com

SILVER:



AGRIS
ADVANCING HORTICULTURE

Η αγροτική
Moonlight
VILLAS



ALAGRO
ALTERNATIVE AGRICULTURAL OPTICS

ΠΙΕΤΡΑΚΗΣ
ΟΙΚΟΔΟΜΙΚΕΣ ΕΡΓΑΣΙΕΣ
BUILDING CONSTRUCTION WORK



NALU
A PARKED CONCEPT

CREaTE Constructions
DESIGN BUILD LIVE

BRONZE:



SUPPORTERS:

